PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED

Topic : Chemical Equilibrium & Ionic Equilibrium (Question Bank # 01)

M.M. : 68 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C. The concentration of H^+ at equivalence point is ($K_w = 1 \times 10^{-14}$ at 25°C)

(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

2. How many moles of NH_3 must be added to 2.0 litre of 0.80 M $AgNO_3$ in order to reduce the Ag^+ concentration to 5×10^{-8} M. K_f of $[Ag(NH_3)_2^+] = 10^8$

(A) 0.4 (B) 2 (C) 3.52 (D) 4

3. Let solubility of a sparingly soluble salt of strong base and weak acid in pure water is equal to s_1, in a buffer solution with pH < 7 is equal to s_2 and in another buffer solution with pH > 7 is equal to s_3. If hydrolysis is considered in all the solutions then the correct relation between s_1, s_2 and s_3 is

(A) $s_2 = s_2 = s_3$ (B) $s_1 > s_2 > s_3$ (C) $s_1 < s_2 < s_3$ (D) $s_2 > s_1 > s_3$

4. For equilibrium at 300 K, $LiCl.3NH_3(s) \rightleftharpoons LiCl.NH_3(s) + 2NH_3(g)$; $Kp = 9$ atm^2

In 8.21 L vessel 1.4 mol of $LiCl.NH_3(s)$ and 2.8 moles of NH_3 are placed. Moles of $LiCl.3NH_3(s)$ formed at equilibrium will be :

(A) 1.4 (B) 0.7 (C) 0.9 (D) 1

5. Two moles of an equimolar mixture of two alcohols R_1–OH and R_2–OH are esterified with one mole of acetic acid. If only 80% of the acid is consumed till equilibrium and the quantities of ester formed under equilibrium are in the ratio 3 : 2. What is the value of equilibrium constant for the esterification of R_1–OH?

(A) 0.48 (B) 3.69 (C) 2.2 (D) 0.75

6. An amount of 1 mole each of A and D is introduced in 1L container. Simultaneously the following two equilibria are established.

$A \rightleftharpoons B + C; K_C = 10^6$ M and $B + D \rightleftharpoons A; K_C = 10^{-6}$ M^{-1}

The equilibrium concentration of A will be

(A) 10^{-6} M (B) 10^{-3} M (C) 10^{-12} M (D) 10^{-4} M

7. Assuming no change in volume, calculate the minimum mass of NaCl necessary to dissolve 0.010 mol AgCl in 100 L solution. [$K_f(AgCl_2^-) = 3 \times 10^5$, $K_{sp} = (AgCl) = 1 \times 10^{-10}$]

(A) $\frac{10}{3}$gm (B) 195 gm (C) 19.5 kg (D) 1950 gm

8. The degree of hydrolysis of a salt of weak acid and weak base in it's 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be

(A) 100% (B) 50% (C) 25% (D) none of these

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(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

9. Which of the following is true for alkaline aqueous solution?

(A) $pH > \frac{pk_w}{2}$ (B) $pH > pOH$ (C) $pOH < \frac{pk_w}{2}$ (D) $pH < pOH$

10. An amount of 1 mole of $PCl_3(g)$ and 1 mole of $PCl_5(g)$ is taken in a vessel of 10 L capacity maintained at 400 K. At equilibrium, the moles of Cl_2 is found to be 0.004

(A) K_C for the reaction ; $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 0.0004 M

(B) K_p for the reaction ; $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$ is $0.0004 \times (0.0821 \times 400)$ atm.

(C) If $PCl_3(g)$ is added to the equilibrium mixture, ΔG at the new equlibrium becomes greater than the ΔG at old equilibrium

(D) After equilibrium is achieved, moles of PCl_3 are doubled and moles of Cl_2 are halved simultaneously, then the partial pressure of PCl_5 remains unchanged.

11. At 90°C pure water has $[H^+] = 10^{-6}$ M. Based on this correct statement is (are) -

(A) Increasing the temperature, increases degree of ionization of water.

(B) At 60°C, pH of pure water will be less than 6

(C) At 90°C, pH of a 10^{-6} M HCl solution will be less than 6

(D) Ionization constant (K_a) of H_2O at 90°C is 10^{-12}

12. Select the correct statement(s) :

(A) If pOH of asolution at 70ºC is 7, then solution will be acidic

(B) If $[H^+]$ concentration is decreases 100 times, pH of solution increases by 2 units

(C) If solution of CH_3COONa(a) is diluted then pH increases.

(D) Buffer capacity remains constant with dilution.

13. Aqueous solutions of HNO_3, KOH, CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) –

(A) HNO_3 and CH_3COOH (B) KOH and CH_3COONa

(C) HNO_3 and CH_3COONa (D) CH_3COOH and CH_3COONa

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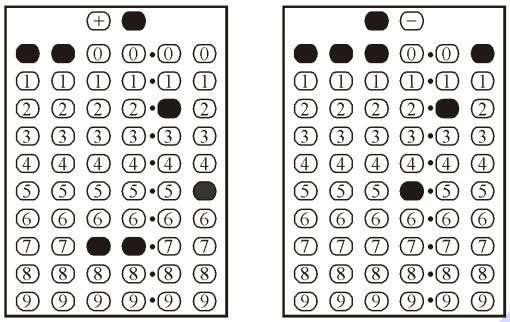
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SECTION-II: (Integer Second Decimal)

- This section contains **4** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If ONLY the correct numerical value is entered as answer.

  Zero Marks : 0 In all other cases.

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1. For the reaction : $A(g) \rightleftharpoons 2B(g)$, the equilibrium partial pressures of gases A and B is 4 bar and 2 bar, respectively. If the equilibrium mixture is slowly & isothermally compressed to 12 bar, maintaining equilibrium, then the new equilibrium pressure(in bar) of B(g) becomes.

2. A weak base BOH was titrated against a strong acid . The pH at 1/4th equivalence point was 8.0. Enough strong base was now added (7.5 millimole) to completely convert the salt. The total volume was 10 ml. Find the pOH at this point.

3. How much water (in L) should be added to 10 g CH_3COOH to give pOH equal to 11.

$K_a(CH_3COOH) = 1.8 \times 10^{-5}$

4. If $[H^+]$ of a 1 litre solution made by mixing 0.01 mol HCl with 1 mol BCl is xM, then the value of '1000x' is.

[Given : pK_b of BOH = 10, $\sqrt{5} = 2.236$]

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SECTION-IV: (Matrix Match type)

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1.

Column-I	**Column-II**
(A) $2AB_{3(g)} \rightleftharpoons A_{2(g)} + 3B_{2(g)}$	(P) Introduction of $B_{2(g)}$ at constant volume shift equilibrium in backward direction
(B) $AB_{3(g)} \rightleftharpoons AB_{(g)} + B_{2(g)}$	(Q) α increases on increasing temperature
(C) $2AB_{(g)} \rightleftharpoons A_{2(g)} + B_{2(g)}$	(R) On adding inert gas at constant pressure equilibrium concentration will change
(D) $AB_{2(s)} \rightleftharpoons A_{(g)} + B_{2(g)}$	(S) On increasing the pressure, concentration of all reactants increases
	(T) α is independent of equilibrium pressure.

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ANSWER KEY

SECTION–I : (i) Only One option correct Type: 3(–1)

1.	**Ans.(D)**	**2.**	**Ans.(D)**	**3.**	**Ans.(D)**	**4.**	**Ans.(C)**
5.	**Ans.(B)**	**6.**	**Ans.(A)**	**7.**	**Ans. (C)**	**8.**	**Ans. (B)**

(ii) One or more options correct Type : 4(–1)

9	**Ans.(A, B, C)**	**10.**	**Ans.(A,D)**	**11.**	**Ans.(A,C)**	**12.**	**Ans.(A,B)**
13.	**Ans. (C, D)**						

SECTION-II: (Integer Second Decimal) : 4(0)

1.	**Ans.(3.00)**	**2.**	**Ans.(3.00)**	**3.**	**Ans. (2.94 to 3.00)**
4.	**Ans.(16.10 to 16.30)**				

SECTION–IV : Matrix-Match Type : 8(0)

1. **Ans.(A)-P,Q,R,S ; (B) P, Q, R, S, (C)- P, Q, R, S,T, (D)-P,Q**

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